Expressions, Equations & Volume: Postorder Traversal & Cubic Units

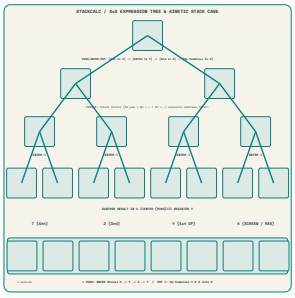
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.5.OA.A.1

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Tree Board

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Assemble cubic volume model: base 4×5 cubes, height 3 layers. Total 60 cubes.

Sketch Model on Paper:
Draw an expression tree with root operator [×] and leaf operands.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	4	X: 4	Leaf operand 4 into
2.	ENTER	Y: 4 | X: 4	Stack lift
3.	6	Y: 4 | X: 6	Leaf operand 6 into
4.	+	X: 10	Execute inner additi

EXPECTED LCD X: 70

1 **Evaluate (18 - 8) × 6 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Evaluate 30 + (8 × 5) on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Find volume of prism: length 7, width 3, height 4.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Find volume of cube with edge length 4 cm.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 1

Expressions, Equations & Volume: Postorder Traversal & Cubic Units • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: 4x4 Expression Tree Case & Cubic Blocks

CAD Blueprint: tree_board

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/build/scad/tree_board.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Expression Tree Tray with White Tokens (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Evaluate (18 - 8) × 6 on the stack.** Standard: CCSS.MATH.5.OA.A.1	60	[1][8] [ENTER] [8] [-] [6] [×]
2	**Evaluate 30 + (8 × 5) on the stack.** Standard: CCSS.MATH.5.OA.A.1	70	[8] [ENTER] [5] [×] [3][0] [+]
3	**Find volume of prism: length 7, width 3, hei** Standard: CCSS.MATH.5.OA.A.1	84 cubic units	[7] [ENTER] [3] [×] [4] [×]
4	**Find volume of cube with edge length 4 cm.** Standard: CCSS.MATH.5.OA.A.1	64 cm³	[4] [ENTER] [4] [×] [4] [×]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

No parentheses required in RPN: inner operations evaluate first automatically.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.